Minwoo Kang, Ph.D.

Postdoctoral Scholar · HR Thiam Lab · Department of Bioengineering, Stanford University
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RESEARCH INTERESTS

Cellular and nuclear mechanobiology, immune-cell and cancer-cell biophysics, quantitative live-cell imaging, biomedical image analysis, and AI-assisted cellular phenotyping

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Feb. 2018–Feb. 2023
Department of Mechanical Engineering — Ph.D. Daejeon, Republic of Korea
Advisor: Prof. Jennifer H. Shin (Department of Mechanical Engineering, KAIST)
Dissertation: Mechanical Perspectives on Fibroblasts in Wound Healing and Tumorigenesis

Soongsil University Mar. 2015–Feb. 2017
Department of Mechanical Engineering — M.S. Seoul, Republic of Korea
Advisor: Prof. Sang-Ho Suh (Department of Mechanical Engineering, Soongsil University)
Thesis: Study on Geometry Improvement of Francis Turbine Runner to Reduce the Sediment Erosion

Soongsil University Mar. 2009–Feb. 2015
Department of Mechanical Engineering — B.S. Seoul, Republic of Korea
Advisor: Prof. Sang-Ho Suh (Department of Mechanical Engineering, Soongsil University)
Thesis: Study of Valid Inlet Boundary Condition of Upper Airway Analysis considering Respiration

HONORS AND AWARDS

CRI Irvington Postdoctoral Fellowship 2026–2029
Cancer Research Institute (CRI), **US$243,000 total fellowship support**

Project: Hijacking an innate immune enzyme: tumor-intrinsic PAD4 remodels nuclear mechanics to promote breast cancer invasion

Stanford School of Medicine Dean's Postdoctoral Fellowship 2025
Stanford University, **US$36,900 total fellowship support**

Project: How does chromatin exit the nucleus to kill pathogens/damage the host during NETosis?

Best Paper Award 2023
Korean Society for Precision Engineering

Venture Research Program for Master's and PhD Students in the College of Engineering 2019
KAIST, **₩9,000,000 research grant**

Project: Hox gene as a regulator for mechanosensitive pathways

Gold Prize 2014
4th Soongsil Engineering Award

Excellence Award (3rd Prize) 2014
National College Student Fluid Engineering Competition, Annual Meeting of Fluid Engineering Division, KSME

Excellence Award (3rd Prize) 2014
4th Capstone Design Award, Soongsil University

RESEARCH EXPERIENCE

Postdoctoral Scholar, Department of Bioengineering, Stanford University
Stanford, CA, U.S. · Oct. 2023–Present
Advisor: Prof. Hawa Racine Thiam (Departments of Bioengineering, Microbiology & Immunology, Stanford University; Sarafan ChEM-H Institute Scholar; Esther Ehrman Lazard Faculty Scholar; Biohub Investigator)

- **Biophysical mechanism of nuclear envelope rupture during NETosis**
  - Measuring nuclear envelope rupture force and intranuclear pressure of neutrophils undergoing NETosis with AFM
  - Mathematical modeling of nuclear envelope rupture

Guest Researcher, National Institute of Biomedical Imaging and Bioengineering (NIBIB), NIH
Bethesda, MD, U.S. · Jul. 2024
Advisor: Alexander Cartagena-Rivera, Ph.D. (Earl Stadtman Investigator, NIBIB, NIH)

Graduate Research Assistant, Department of Mechanical Engineering, KAIST
Daejeon, Republic of Korea · Feb. 2018–Feb. 2023

- **Cell Mechanics based Analysis of Scar Formation for Strategic Scar Treatment (NRF-2017R1A2B2007673, 2017–2020)**
  - **Led** experimental design, execution, and data analysis
  - Developed a **patented multi-cell tensile stimulation device**
  - Research outputs included **one first-author and one co-authored peer-reviewed article**
- **AI-based classification of phenotype and function of cancer-associated fibroblasts for novel therapeutic strategy in pancreatic cancer (NRF-2021R1A2C3008408, 2021–2025)**
  - **Conceived the project and co-wrote the grant proposal**
  - Developed a deep-learning pipeline for quantitative cellular phenotyping and contributed to **a patented hybrid-learning method for cell-subtype classification**
  - Research outputs included **two first-author and two co-authored peer-reviewed articles**
- **Development of 3D biopatch-applied breast implant technology with anti-stress micro patterns and drug delivery system for the prevention of capsular contracture (22B0401L1, 2022–2025)**
  - Supported writing the grant proposal
  - Classified fibroblast subtypes from different breast regions using biophysical features and machine-learning algorithms

PUBLICATIONS

Journal Articles & Preprints

1. **Kang, M.**, Cabral, A. T., Sawant, M., & Thiam, H. R. (2026). Benchmarking three simple DNA staining-based image metrics for live-cell tracking of chromatin organization. ***Nucleus.*** *(accepted)*
2. Cabral, A. T., Sawant, M., **Kang, M.**, Xie, L., & Thiam, H. R. (2026). Chromatin decompaction within the nucleus increases plasma membrane tension promoting NETosis execution, independently of transcription. ***Nature Communications***. https://doi.org/10.1038/s41467-026-76578-1
3. **Kang, M.**, Devarasou, S., Kwon, T. Y., & Shin, J. H. (2025). EMT induction in normal breast epithelial cells by *COX2*-expressing fibroblasts. ***Cell Communication and Signaling***, 23(1). https://doi.org/10.1186/s12964-025-02227-7
4. **Kang, M.**, Min, C., Devarasou, S., & Shin, J. H. (2025). AI-driven classification of cancer-associated fibroblasts using morphodynamic and motile features. ***APL Bioengineering***. https://doi.org/10.1063/5.0250502 ***(selected as a featured article)***

5. **Kang, M.**, Ko, U. H., Oh, E. J., Kim, H. M., Chung, H. Y., & Shin, J. H. (2025). Tension-sensitive HOX gene expression in fibroblasts for differential scar formation. ***Journal of Translational Medicine***, 23(1), 168. https://doi.org/10.1186/s12967-025-06191-1
6. Devarasou, S., **Kang, M.**, & Shin, J. H. (2024). Biophysical perspectives to understanding cancer-associated fibroblasts. ***APL Bioengineering***, 8(2). https://pubs.aip.org/aip/apb/article/8/2/021507/3296522
7. Devarasou, S., **Kang, M.**, Kwon, T. Y., Cho, Y., & Shin, J. H. (2023). Fibrous matrix architecture-dependent activation of fibroblasts with a cancer-associated fibroblast-like phenotype. ***ACS Biomaterials Science & Engineering***, 9(1), 280–291. https://doi.org/10.1021/acsbiomaterials.2c00694
8. Lee, J. S., Cho, H. G., Lee, J. W., Oh, E. J., Kim, H. M., Ko, U. H., **Kang, M.**, Shin, J. H., & Chung, H. Y. (2023). Influence of transforming growth factors beta 1 and beta 3 in the scar formation process. ***Journal of Craniofacial Surgery***, 34(3), 904–909.
9. Park, N., Kim, K., **Kang, M.**, Suh, S.-H., Ryu, G., Kim, S. J., & Park, T. (2017). A comparative study on the Francis turbine performances according to the different vane shape. ***The KSFM Journal of Fluid Machinery***, 20(4), 19–24.
10. Kim, H.-H., Jung, Y. H., **Kang, M.**, & Suh, S.-H. (2017). A comparative study on the operation capabilities according to the different capsule shapes in pneumatic capsule pipeline. ***The KSFM Journal of Fluid Machinery***, 20(4), 56–61.
11. **Kang, M.**, Park, N., & Suh, S.-H. (2016). Numerical study on sediment erosion of Francis turbine with different operating conditions and sediment inflow rates. ***Procedia Engineering***, 157, 457–464.

Selected International Conference Presentations

- **Kang, M.**, Cartagena-Rivera, A., & Thiam, H. R. “PAD4-dependent chromatin decompaction and nuclear envelope remodeling drive explosive nuclear rupture during NETosis.” *Biophysical Society*, Feb. 2026.
- **Kang, M.**, Devsrasou, S., Min, C., & Shin, J. H. “Classification of cancer-associated fibroblasts with morphodynamics and motility feature-based AI platform.” *International Conference on Precision Engineering and Sustainable Manufacturing*, Jul. 2023. **(Best Paper Award)**
- **Kang, M.**, Ko, U. H., Oh, E. J., Lee, J. W., Chung, H. Y., & Shin, J. H. “*HOX*-driven mechanobiology of scar formation.” *IUPESM WC*, Jun. 2022.
- **Kang, M.**, Ko, U. H., Oh, E. J., Lee, J. W., Chung, H. Y., & Shin, J. H. “Loss of tensile stress in skin wound regulates scar formation.” ***Invited talk for Korea Satellite Meeting*** — Materials, Mimics, and Microfluidics: Engineering Tools for Mechanobiology (MBI 3M), Jul. 2021.

Patents

1. Jennifer H. Shin, Chanhong Min, **Minwoo Kang**, Hyun Tae Jeong, Tae Yoon Kwon, & Myung Chul Kim. “A method and apparatus for classification of subtypes of cells with morphological and motility features using hybrid learning.” Jan. 2025 (Reg. No. 1027625420000, Republic of Korea).
2. Jennifer H. Shin, **Minwoo Kang**, Ung Hyun Ko, & Hyun Tae Jeong. “Multi-cell stimulation device.” Dec. 2021 (Reg. No. 1023410530000, Republic of Korea).

MENTORING EXPERIENCE

Mentored a junior lab member to win a student research grant — “Deep learning-based quantitative evaluation indicators for the neural differentiation,” *Venture Research Program for Master's and PhD Students in the College of Engineering* May 2022

Mentored a junior lab member to win a student research grant — “AI-based morphological classification of heterogeneous populations in breast cancer cells,” *Venture Research Program for Master's and PhD Students in the College of Engineering* May 2021

Master's program research mentor — “The effect of 3D fibrous gelatin network property in activation of cancer-associated fibroblasts (CAFs) in breast cancer cells,” *Master's thesis, KAIST* Mar. 2019–Feb. 2021

SKILLS

Computational — Python-based quantitative image analysis and computer vision for microscopy, including image segmentation, cell tracking, feature extraction, and machine/deep learning for time-series cellular data, Fiji/ImageJ, COMSOL Multiphysics, ANSYS-CFX, ANSYS Mechanical, ICEM-CFD

Experimental — Basic cell and spheroid culture, Live-cell microscopy, Fluorescent (confocal and super resolution) microscopy, Light-sheet microscopy, Cell traction force and monolayer stress microscopy, Atomic force microscopy (AFM), Scanning electron microscopy (SEM), Collective cell micropatterning, Hydrogel fabrication, Soft-lithography fabrication, Electrospinning, Tensile stimulation apparatus development, Quantitative reverse-transcription PCR (RT-qPCR), Transfection

Languages — Korean (native), English (professional working proficiency)

MILITARY SERVICE

Discharged as a sergeant, Republic of Korea Army · Incheon, Republic of Korea Mar. 2011–Dec. 2012

REFERENCES

Prof. Hawa Racine Thiam *(Postdoctoral advisor)*
Departments of Bioengineering, Microbiology & Immunology, Stanford University
hrthiam@stanford.edu

Prof. Jennifer H. Shin *(Ph.D. program advisor)*
Department of Mechanical Engineering, KAIST
j_shin@kaist.ac.kr

Prof. Bomi Gweon *(Ph.D. dissertation committee member)*
Department of Mechanical Engineering, Sejong University
bgweon@sejong.ac.kr

Prof. Hyoung-Ho Kim *(B.S. and M.S. program mentor)*
School of Mechanical Material Convergence Engineering, Gyeongsang National University
khh106@gnu.ac.kr